

10

Acquisition of the syntax–semantics interface

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10.1 Types of meaning

Few people start learning a second language for the exotic sounds, or for the elegant sentence structure that they detect in it. *Meaning* is what we are all after. We would all like to understand and to be able to convey thoughts and feelings and observations in another language the way we do in our native language. Ever since Aristotle, linguists have considered language to be the pairing of form (sound or written strings) and meaning. In this chapter, we examine the road to meaning, that is, how we come to understand and convey meaning in a second language, and where the pitfalls to that may lie.

We will begin by distinguishing between several types of meanings that go in a linguistic message (although some of this will be reinforcement of material from previous chapters):

- lexical
- (phrasal) semantic
- grammatical
- discourse-related
- pragmatic.

For many people, when they think of learning a foreign language, semantics describes predominantly what meanings are encoded in the foreign words. These are lexical meanings. For example, the English *cat* is *gatto* in Italian; both words denote “a small furry animal.” The term “denotation” stands for the literal, or primary meaning of a word. It is common to think of these meanings as stored in our mental lexicon and being accessed whenever we need them to use in a sentence we want to produce.

Phrasal, or sentential, semantics is compositional, that is, it is calculated by combining the meanings of all the words in a sentence and taking their order and other rules into account. Take for example the English sentence in (1).

- (1) Cats were exterminated in a cataclysm.

Think about the word “cats” in this sentence as if you hear this sentence as part of a story. What does the word mean? Depending on the context, it may mean that *a number of cats* were exterminated in a specific tragic event in the past, but it can also mean (untruthfully) that *all cats* were affected by a cataclysm and are now extinct like dinosaurs. You can think of stories that would make the two different meanings plausible. Of these two meanings of *cats*, only the first one is available for the equivalent Italian sentence while the second is not (Longobardi 2001).

- (2) Gatti sono stati sterminati da un cataclisma.
cats were exterminated in a cataclysm.
First meaning: Some (unspecified) number of cats
Second meaning: Cats in general, all cats
(not available in Italian, available in English)

Although *cats* and *gatti* have the same lexical meaning in both languages, when used in a sentence, they give rise to two different meanings in English (*some cats*, *all cats*), only one meaning in Italian (*some cats*). This difference

is captured and explained by the rules for calculating sentence meaning in the two languages, combined with some syntactic rules, and is the research focus of (phrasal) semantics.

Grammatical meaning also comes into consideration in calculating sentential meaning. Consider the two sentences below:

(3) Jane eats sushi.

(4) Jane ate sushi.

They contain two identical lexical items (*Jane*, *sushi*) and the third, the verbal form, encodes a grammatical difference in tense and aspect. We understand that a present habitual (but not an ongoing) event is meant by the utterance in (3) while a past habitual event or a past completed event is a possible reading of (4). As we saw in Chapter 7, grammatical meanings are mostly encoded in inflectional morphology, for example *-ed* for past simple, *-s* for 3rd person singular present simple, etc.

A fourth type of meaning depends on context consideration. It is well known that the neutral English word order is Subject–Verb–Object, as for example in (3). So what are the fronted objects in a sentence like (5) signifying?

(5) Question: Does Jane like sushi?

Answer: Sushi, she eats; but sashimi, she adores.

When the object is already mentioned in the discourse, English can front it (move the object before the verb) in a context-dependent operation called “topicalization.” The meaning of this fronting is something along the lines of: “If we’re talking about sushi, Jane eats it, . . .” The sentence is rendered even more natural by the mentioning of another fronted object, sashimi, in a contrast between the two. Note that the meaning of topic depends on the previous sentence, so it is discourse-related.

Finally, there are sentences in which what a speaker says can fail to be fully meaningful without considering an additional implication: pragmatic meaning. The latter is the object of investigation of linguistic pragmatics. Scalar implicatures are a good example of the dissociation between semantic (or logical) and pragmatic meaning. Consider the following example of a well-known pragmatic inference. When we hear the sentence in (6a) we

actually understand that the speaker wants to say something stronger along the lines of (6b).

- (6) a. Bonnie ate some of the strawberries.
b. Bonnie did not eat all of the strawberries.

Notice that the meaning *not all* is not encoded by the speaker's utterance, nor is it part of what the speaker has said. Rather, that interpretation is an assumption inferentially derived by the hearer on the basis of what the speaker has said. Logically speaking, *some* means *some and possibly all*. For pragmatic felicity, however, *some* means *some but not all*. The rationale goes like this: If the speaker wanted to say that *some and possibly all* strawberries are gone, she would have uttered *Bonnie ate (all) the strawberries*, being maximally informative. Since she did not, she must really mean that there are some strawberries left. Understatements of this sort in human speech are regulated by Gricean maxims, and more specifically, the Maxim of Quantity: Make your contribution as informative as is required; do not make your contribution more informative than is required (Grice 1989). Lexical items that induce such calculations are arranged on a scale: *<some, most, all>*, *<start, finish>*, and so on, where uttering the lower placed item implies that the higher placed item is not true.

Indexical lexical items such as *today*, *this month*, etc. provide another example of pragmatic meaning. It is common knowledge that the denotation of the word *today* changes every day, and so the sentence in (7) will be true if uttered on a Tuesday but false if uttered the next day. Such pragmatic knowledge is truly universal.

- (7) Today is Tuesday.

The boundaries between semantic and pragmatic meaning are notoriously fluid, and debated by many linguists. A rule of thumb is that semantic, compositional meaning is calculated disregarding the context and general knowledge of the world. For example, let us consider the cat example one more time. An English speaker who is able to calculate the two compositional meanings in (1) and compare them will also be able to discard the second one as untrue if this person knows that cats do exist in the present day. Thus, the first meaning will be preferred, unless some special context is invoked (such as a fairy tale, or science fiction). That second process

(checking the plausibility of an available interpretation against knowledge of the world and the discourse situation) is a pragmatic calculation. Semantics proposes, pragmatics disposes.¹

Let us further illustrate the various linguistic meanings with the different types of information that is brought from the lexicon, syntax, semantics, and pragmatics and that lead to the calculation of meaning of the sentence in (8).

(8) Mona likes tea.

An important characteristic of the verb *like* is that its subject is someone (a sentient being) who experiences a feeling, let's call it the Experiencer. Its object is the person, event, or thing that gives rise to the feeling of liking, or a Theme. The noun *tea* in the lexicon is ambiguous between a drink and the dried tea leaves from which the drink is made (think of these two meanings available in the sentence *Where is the tea?*).² However, the sentence in (8) is not really ambiguous, since most hearers would interpret *tea* as the drink. That is, out of the two possible lexical meanings, one is brought forward by the sentence in (8). The agreement marker *-s* carries the grammatical meanings of present tense and habitual aspect (among others). Thus, we know that the feeling of liking is characteristic of Mona in a time period including the present moment. The word order in (8) is neutral. This fact in itself carries information about the type of discourse where the sentence is felicitous, especially when we compare it to the topicalized sentence *Tea, Mona likes* or the focused *TEA Mona likes, not coffee*. Finally, knowledge of pragmatics tells us that the full meaning of the sentence *Mona likes tea* is most likely *Mona likes to drink tea*, as opposed to watching it or driving it. The latter would be unusual or inappropriate actions one does with a drink.

¹ The following example, borrowed from David Lodge's *Paradise News*, reinforces the latter point:

'I just met the old Irishman and his son, coming out of the toilet.'

'I wouldn't have thought there was room for the two of them.'

'No silly, I mean *I* was coming out of the toilet. They were waiting.' (1992: 65).

² In some parts of the United Kingdom, "tea" can also have a third meaning: "an evening meal."

Practice: Before going any further, try to distinguish the various meanings (lexical, grammatical, semantic, discourse-related, and pragmatic) in the following sentences:

1. The writer began the book.
2. The goat began the book.

10.2 Mismatches at the syntax–semantics interface

All meanings can be conveyed in all languages. This is a notion worth repeating because it is the cornerstone of our approach to second language meaning. All languages can convey all meanings. A concept such as *liking* with its participants the Experiencer and the Theme, grammatical meanings such as present time and habitual action, pragmatic meanings such as topic and focus, and implied meaning are all universal. However, languages may have different rules for encoding these meanings and for putting them together in various morphemes, words, phrases, and word orders. When universal grammatical meanings are captured by different grammatical morphemes, as we shall see in the examples below, it is not enough to just learn the new words and put together the L2 sentences. L2 learners have to do some special restructuring in their grammar.

In this section, we will go through several examples of syntax–semantics mismatches and the studies that investigated the L2 acquisition of these properties. While the content of meaning is the same (concepts and relations between them), different linguistic forms map different natural groupings of meanings. Let us start by illustrating a mismatch at this interface with Spanish and English aspectual tenses. While the English past progressive tense signifies an ongoing event in the past, Spanish Imperfect can have both an ongoing and a habitual interpretation. The English simple past tense, on the other hand, has a one-time completed event interpretation and a habitual interpretation while the Spanish Preterit has only the former.

(9) a. *Habitual event*

Guillermo robaba en la calle.

Guillermo rob-IMP in the street

‘Guillermo habitually robbed (people) in the street.’

- b. *One-time completed event*
 Guillermo robó en la calle.
 Guillermo rob-PRET in the street
 ‘Guillermo robbed (someone) in the street.’
- c. *Ongoing event*
 Guillermo robaba a alguien en la calle (cuando llegó la policía)
 Guillermo rob-IMP someone in the street (when arrived the police)
 ‘Guillermo was robbing someone in the street when the police arrived.’
- (10) a. *Habitual event*
 Felix robbed (people) in the street.
- b. *One-time completed event*
 Felix robbed a person in the street.
- c. *Ongoing event*
 Felix was robbing a person in the street (when the police arrived).

Thus, the same semantic primitive meanings (ongoing, habitual, and one-time finished event), arguably part of universal conceptual structure, are distributed over different pieces of functional morphology in English and in Spanish. Table 10.1 illustrates the mismatch.

Another example comes from the marking of politeness in languages like German, Bulgarian, and Russian, as opposed to English. German uses the second person plural pronoun *Sie* in situations when the addressee is singular, but the speaker wants to be polite, and employs *du* (you-SG) for all other cases. Bulgarian uses *Vie* ‘you-PL’ and *ti* ‘you-SG’ in exactly the same way. English does not reflect this distinction in the morphology of personal pronouns. Again, that does not mean that English speakers have no concept of politeness. They just express it with other means, for example the polite *Sir*, *Madam*, or *Miss*.

Table 10.1 Mapping between forms and meanings in Spanish and English aspectual tenses

	One-time completed event	Habitual event	Ongoing event
English	past simple		past progressive
Spanish	Preterit	Imperfect	

When learning a second language, speakers are faced with four different acquisition tasks regarding meaning: they have to learn the lexical items of the target language, that is, map linguistic form and lexical meaning one by one. This is certainly a laborious task. Learning the functional morphology is qualitatively different: abstracting away from irregular morphology, once a speaker learns that *-ed* in English encodes a past habit or a past completed event, she can apply this knowledge to all English regular verbs. Sentential and pragmatic meanings are calculated using universal mechanisms of human language. Once the lexical and grammatical meanings are learned, sentential and pragmatic meanings come for free and do not constitute a barrier for acquisition. However, the hardest acquisition situation is when there is a mismatch between which meanings are mapped onto which forms in the native and the second language. In what follows, we will look at some studies investigating such learning situations.

10.3 L2 acquisition of syntax–semantics mismatches

The linguistic properties whose acquisition Slabakova (2003) investigates have to do with grammatical aspect: meanings like ongoing or completed event exemplified in the section above. English differs from German, Romance, and Slavic with respect to the semantics of the present tense. It is well known that the English bare infinitive denotes not only the process part of an event but includes the completion of that event. When talking about *crossing the street*, for example, there is a part of the event which is a process, the walking, and another part which is the completion, taking the last step having crossed the street.

- (11) a. I saw Mary cross the street. (completion entailed)
b. I saw Mary crossing the street. (no completion entailed)

In trying to explain the facts illustrated in (11), many researchers have noticed that English verbal morphology is impoverished. The experimental study adopts Giorgi and Pianesi's (1997) proposal. English verbs such as *cross*, they argue, are "naked" forms that can express several verbal values, such as the bare infinitive, the first and second person singular, and the first, second, and third person plural. Some verbal forms are indistinguishable from nouns. Giorgi and Pianesi (1997) propose that verbs are distinguished in English by

being marked in the lexicon with the aspectual feature [+perfective], standing for ‘completed.’ Thus, children acquiring English can differentiate verbal forms from nouns, since the latter will not have the feature [+perfective].

In Romance, Slavic, and other Germanic languages, on the other hand, all verbal forms have to be inflected for person, number, and tense. Thus, nouns and verbs cannot have identical forms, unlike in English. The Bulgarian verb, for example, is associated with typical verbal inflections, such as person and number, and it is recognizable and learnable as a verb because of these inflections. Bulgarian verbs are therefore not associated with a [+perfective] feature. Consequently, Bulgarian equivalents to bare infinitives do not denote a complete event, as (12) illustrates.

- (12) Ivan vidja Maria da presiča ulicata. (no completion entailed)
 Ivan saw Maria to cross street-DET
 ‘John saw Mary crossing the street.’

In the acquisition of English by Bulgarian native speakers, then, the learning task is to notice the fact that English inflectional morphology is highly impoverished, lacking many person–number–tense verb endings.

However, this is not all. The attachment of the feature [+perfective] to English eventive verbs would bring along, if Giorgi and Pianesi are correct, knowledge of four different interpretive facts: 1) bare verb forms denote a completed event (as in 11a); 2) present tense has only habitual interpretation (*Jane goes to school in the morning*); 3) the progressive affix is needed for ongoing interpretation of eventive verbs (*Jane is going to school right now*); 4) states in the progressive denote temporary states (*Jane is being stupid*). In contrast, Bulgarian has no progressive tense, and the present simple tense is ambiguous between habitual and ongoing events, characteristic or temporary states.

Practice: Before going any further, discuss these four meanings with other students. You can use these examples or give some others:

1. I heard Mary sing a German aria.
2. Monty repairs cars.
3. Mary is repairing her car.
4. Marilyn is being lazy.
5. Marilyn is lazy.

Which sentences do you predict will be the same in Bulgarian?

All the properties above follow from (or are related to) the presence of the feature [+perfective] on the English verb. Thus the syntax-semantics mismatch is really much wider: a minimal difference between languages—the presence or absence of a feature in the lexicon—leads to various and superficially unconnected interpretive mismatches. All of the meanings that have to be learned are not attested in the native language of the learners. Even more importantly, of the four meanings enumerated above, the second, third, and fourth are introduced, discussed, and drilled in language classrooms. The first one, however, is not explicitly taught.

Let us consider the experimental study now. A hundred and twelve Bulgarian learners of English took part in the experiment, as well as 24 native speaker controls. The learners were divided into groups of low, high-intermediate, and advanced proficiency in English. They were typical classroom-instructed learners. All participants took a production task to make sure they had knowledge of inflectional morphology and a Truth Value Judgment Task with a story in their native language and a test sentence in English. (13) offers an example of a set of four test items. Each story was followed by a single sentence, each story–sentence pair appeared in different places in the test, but I collapse a quadruple here for lack of space. Think about how you would answer if you were taking this test.

- (13) A quadruple testing completed interpretation of English bare verb forms (the construction is known as “perceptual reports”:
X observed/watched/heard Y Verb)

Matt had an enormous appetite. He was one of those people who could eat a whole cake at one sitting. But these days he is much more careful what he eats. For example, yesterday he bought a chocolate and vanilla ice cream cake, but ate only half of it after dinner. I know, because I was there with him.

I observed Matt eat a cake.	True	False
I observed Matt eating a cake.	True	False

Alicia is a thin person, but she has an astounding capacity for eating big quantities of food. Once when I was at her house, she took a whole ice cream cake out of the freezer and ate it all. I almost got sick just watching her.

I watched Alicia eat a cake. True False
 I watched Alicia eating a cake. True False

Results on the acquisition of all four semantic properties pattern the same way. On the three instructed properties (habitual interpretation of the present, progressive needed for ongoing interpretation, states in the progressive denote temporary states), the advanced learners are highly accurate. Intermediate learners are more accurate on the habitual presents than on ongoing progressives. Thus, initial L1 transfer and subsequent morphological acquisition are clearly attested in the data.

Figure 10.1 presents the mean accuracy on the untaught construction, perceptual reports. As we can see in the figure, advanced learners are even more accurate (but not significantly so) than native speakers in their knowledge that an English bare verb denotes a complete event, and consequently is incompatible with an incomplete event story (look at first group of columns). This is a meaning that cannot transfer from the L1, as example (12) indicates.

After establishing that it is *possible* to acquire semantic properties in the second language that are not manifested in the native language, let us now turn to the impact of the instruction variable. Slabakova (2003) reports that

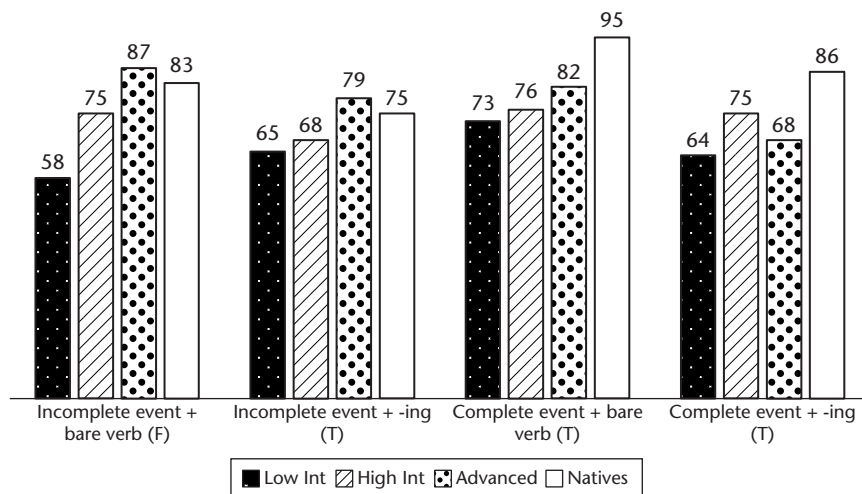


Figure 10.1 Mean accuracy on bare verb versus *-ing* form on perceptual reports in combination with incomplete event or complete event contexts (percent), from Slabakova (2003)

extensive scrutiny of the instruction materials and discussions with the instructors ascertained that the present simple and progressive tense meanings are explicitly taught and drilled from the beginning of classroom instruction. On the other hand, the fact that the bare verb expresses a complete event is not taught, and the Bulgarian teachers are not consciously aware of it. Is it the case that instruction is a significant variable and learners were more accurate on the taught than on untaught properties? The short answer is “no.” Statistical analysis on the data for each group, with condition as the sole factor, indicated that all groups perform equally well on all conditions. The theoretical implication of this finding is that all semantic effects of learning the observable trigger (English verbs are morphologically impoverished) and the related property ([+perfective] feature attached to verbs in the lexicon) appear to be engaged at the same time. Even untaught syntax–semantics mismatches are learnable to a nativelike level.

Another learning situation that exemplifies a syntax–semantics mismatch comes from the work of Ionin, Montrul, and colleagues. In a series of studies, they investigated knowledge of genericity and its expressions in English, Spanish, and Brazilian Portuguese. A noun phrase has generic meaning if it refers to the whole class of individuals denoted by the noun (e.g., *Lions are dangerous animals*). A noun phrase has specific meaning if it refers to something or someone that is known to the speaker, and that is noteworthy in some way in the context (e.g., *A student in Semantics cheated on the exam. His name is David.*) (We shall see specificity contrasted with definiteness later on in this chapter, so you will become experts in identifying specific readings.) Take for example the English and Spanish sentences in (14) through (17), all examples following Ionin, Montrul, and Crivos (2013: 485).

- (14) Tigers eat meat. [√generic reading, *specific reading]
 (15) The tigers eat meat. [*generic reading, √specific reading]
 (16) *Tigres comen carne.
 tigers eat meat
 (17) Los tigres comen carne. [√generic reading, √specific reading]
 the-PL tigers eat meat
 ‘The tigers eat meat.’

The sentence in (14) describes a characteristic of all tigers (the generic meaning) while the equivalent Spanish sentence in (16) is ungrammatical.

Table 10.2 Mapping between forms and meanings in Spanish and English plural NPs in subject position

	Generic meaning	Specific meaning
English	bare plurals	definite plurals
Spanish	definite plurals	definite plurals

The two superficially similar phrases in (15) and (17) also have different meanings: while the Spanish sentence can refer to all tigers in general as well as tigers made specific by a concrete discourse, the English sentence only has the second meaning. Note that this is a complex syntax–semantics mismatch, in which some of the equivalent NPs are unattested, but others are possible with different meanings. Table 10.2 summarizes the form mismatch within the generic meaning. The fact that there is no mismatch within the specific meaning is potentially misleading, because it can lead to overgeneralizations.

Ionin and Montrul (2010) and Ionin, Montrul, and Crivos (2013) studied this contrast in the acquisition of Spanish and English, using a meaning-focused Truth Value Judgment Task and a more metalinguistic acceptability judgment task. Figure 10.2 provides an example from the TVJT with pictures.

Results of this bidirectional study (Spanish to English and English to Spanish) suggest that acquisition can be quite successful, in the sense that learners were influenced by native language transfer but were also able to retreat from it to attain targetlike interpretations. In addition, learning which types of NPs are ungrammatical in the L2 (such as (16)) proved to be easier than learning all the possible NP interpretations, since accuracy with ungrammaticality was documented at earlier levels of proficiency. In this sense, the authors documented a dissociation between narrow syntax and semantic properties, implying that these two types of knowledge may be developmentally uncoupled. Comparing their own results with findings from Serratrice, Sorace, Filiaci, and Baldo (2009), a study using a different methodology,³ the researchers note a similarity between the way bilingual

³ Serratrice et al. (2009) used a form-focused, metalinguistic task, in which the children were asked to make judgments about the grammatical acceptability of the sentences (presented in the context of pictures).

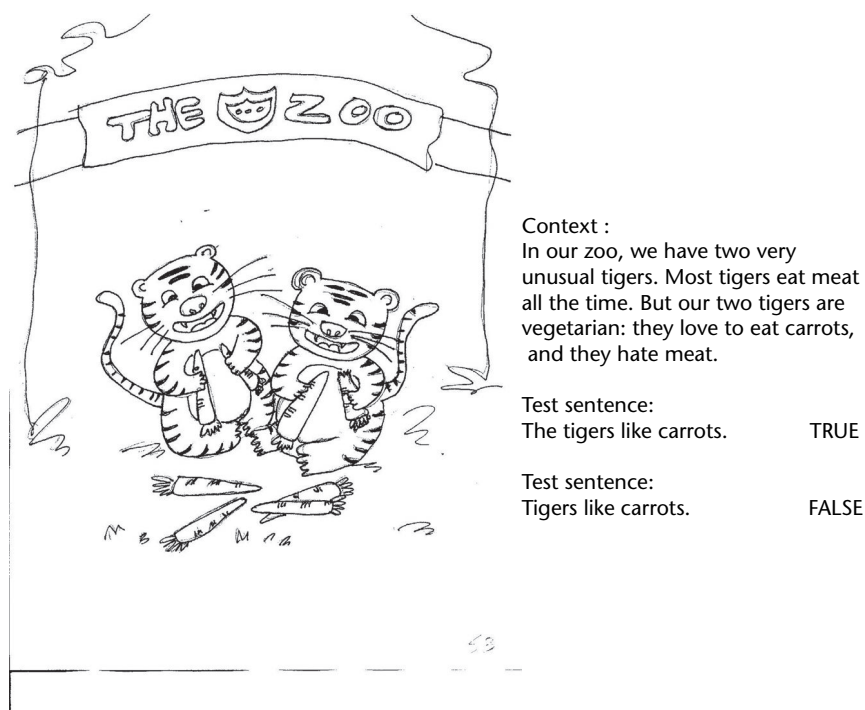


Figure 10.2 Picture and example test items from the Truth Value Judgment Task, with a bare plural and a definite plural test item, from Ionin and Montrul (2010)

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and monolingual children acquire this semantic paradigm and the way adults acquire it. Specifically, all three populations have difficulty deciding which contexts in English are compatible with definite plurals and which with bare plurals, although targetlike behavior comes with increased age and proficiency. However, children and adults have to be tested with the same methods and the same test instrument before any definite conclusions can be drawn about the causes of the learning difficulty.

Working with very advanced learners and looking at the same definite and bare plural noun phrases in subject positions, Cuza, Guijarro-Fuentes, Pires, and Rothman (2013) also document successful L2 Spanish acquisition. Their learners had expanded their grammars to allow definite plurals to have a generic interpretation, but only half of the learners successfully uncoupled bare plurals (ungrammatical in subject position in Spanish)

from the generic interpretation. These two studies point to challenging, although ultimately successful, acquisition of this particular mismatch. Future studies of this and similar complex mismatches could involve fresh language combinations and new form–meaning mappings.

10.4 Poverty of the Stimulus learning situations in semantics

Very close in difficulty to syntax–semantics mismatches are Poverty of the Stimulus (POS) situations. These are learning situations in which the linguistic input to learners is purportedly insufficient for language acquisition to take place. These situations exist in child language acquisition as well, but in L2 acquisition they take on additional significance. In most cases, we are dealing with a learning situation as illustrated in Table 10.3, where two related and minimally different sentences differ in available interpretations as shown. Sentence 1 has two meanings, while one of the meanings is not attested for Sentence 2. However, learners have no way of knowing about the absence of one of the meanings in Sentence 2, without negative evidence. Negative evidence refers to explicit instruction on what is ungrammatical in a language, e.g., a teacher saying to the student, “This is not a good way to say X in language Y.” Since negative evidence of this sort is almost never offered to learners, the assumption is that the innate language faculty aids the learners through constraining their hypothesis space (the hypotheses they entertain about the meaning of a string). This type of research at the syntax–semantics interface was pioneered by Laurent Dekydtspotter and Rex Sprouse (e.g., Dekydtspotter and Sprouse 2001). We will discuss two such studies with variations of the learning tasks in what follows.

A perfect illustration of the learning situation in Table 10.3 is provided by the experimental studies reported in Slabakova and Montrul (2002, 2003) and Montrul and Slabakova (2002, 2003). These studies investigated knowledge of viewpoint aspectual contrasts in Spanish-English interlanguage. One of the conditions in the Truth Value Judgment Task was related to a

Table 10.3 Mapping of sentence strings and meanings in a POS learning situation

	Meaning 1	Meaning 2
Test sentence 1	✓	✓
Test sentence 2	✓	✗!

POS property: the two interpretations of impersonal subjects as illustrated in (18) and (19) below.

- (18) *Se comía bien en casa de la abuela.*
se eat-IMP well in house of the grandmother
'One/We would eat well at grandma's.'
se = la gente en general 'people in general' (generic)
se = nosotros 'we' (specific)
- (19) *Se comió bien en casa de la abuela.*
se eat-PRET well in house of the grandmother
'We ate well at grandma's.'
se = #la gente en general 'people' (#generic)
se = nosotros 'we' (specific)

Notice how similar these sentences are: they differ just in the verbal tense. While the impersonal subject *se* in sentences with the imperfect tense can be interpreted both generically and specifically, the impersonal subject in preterit sentences has only the latter interpretation. Since impersonal subject meanings are not taught in language classrooms, and since knowledge of the absence of one interpretation (out of a possible two) has to be acquired, the authors argued that this is indeed a POS learning situation. They found that advanced learners were over 85% accurate in judging these interpretations, including the lack of one interpretation. The correct knowledge started to emerge even among intermediate learners.

The second study exemplifying a semantic POS, Marsden (2009), looks at the acquisition of distributive quantifiers in L2 Japanese by English and Korean native speakers. Sentences with quantifiers such as *some(one)* and *every(one)* provide the clearest evidence for the fact that some meanings are calculated with movement of the arguments but covert movement, movement that is not visible to the eye. Let's start with English. The sentence in (20) has two possible interpretations.

- (20) Someone read every book.

One interpretation is that there is *some person, call her X, who read every book*. In this construal, we have one person and many books. On the other interpretation, which you may have to work harder to get, there are many

books and many people: *for each book Y, there was some person or other who read that book Y*. Notice that in the paraphrasing of the latter construal, the object has moved (linguists say, taken scope) over the subject. But this movement of the object happens only in our heads, for the purposes of interpretation, while the original sentence stays the same. It just has two possible meanings.

Now, recall the linguistic transformation called “scrambling,” we saw some German scrambled sentences in Chapter 8 (and they reminded us of scrambled eggs). English is averse to scrambling while Japanese and Korean, exemplified below, allow it.

- (21) a. Japanese: Dareka-ga dono hon-mo yonda. (SOV)
 Korean: Nwukwunka-ka enu chayk-ina ilkessta.
 someone-NOM every book read
 ‘Someone read every book.’

Interpretation: There is some person X, such that X read every book.

- b. Japanese: Dono hon-mo dareka-ga yonda. (OSV)
 Korean: Enu chayk-ina nwukwunka-ka ilkessta.
 every book someone-NOM read
 ‘Someone read every book. (scrambled)’

Interpretations: There is some person X, such that X read every book.

For each book Y, some person or other read Y.

In (21a), we have the nonscrambled SOV sentences in Japanese and Korean, and you can notice that they have only one of the two English interpretations. In (21b), the scrambled OSV sentence has both interpretations. Note how this learning situation is very close to the form–meaning mapping illustrated in Table 10.3: two very close strings, produced through a common transformation, scrambling, do not have the same interpretations. The missing interpretation will never be exemplified in input to learners (because it is missing), so how can they learn that the string in (21a) has only one interpretation? This is the Poverty of the Stimulus learning situation that Marsden investigated. In her design, the two learner groups are predicted to behave differently. Korean learners of Japanese should have smooth sailing, since their native language works the same way as the target language. The

English native speakers, on the other hand, will have the hard task of rejecting one interpretation to the canonical word order sentence that their native language makes available. It is a bonus that three native speaker groups (Japanese, Korean, and English) were included in this experimental study, which confirmed the semantics literature claims about these languages' quantifier interactions.

During the experiment, two pictures (as in Figure 10.3) were shown to the participants one after the other for 10 seconds. Then the written sentence was revealed, and at the same time, an audio recording of the sentence was played. "Presenting the picture (that is to say, the interpretation) before the sentence was intended to reduce the possibility of participants determining their own interpretation of each sentence, then rejecting any picture–sentence pairings—possible or not—that did not match their preconceived idea" (Marsden 2009: 144–5).

In the picture on the left in Figure 10.3, one person is stroking every cat, while in the picture on the right, each cat is enjoying the attention of its own stroking person.

The results from the Korean learners confirmed the hypothesis of L1 transfer in the realm of semantics. The judgments of the English native speakers and the English native learners of Japanese were of most interest. Confirming previous findings (Kurtzman and MacDonald 1993), the English native speakers accepted the object wide scope sentences, the second construal of (20), less readily than the first construal, with 67.5% versus

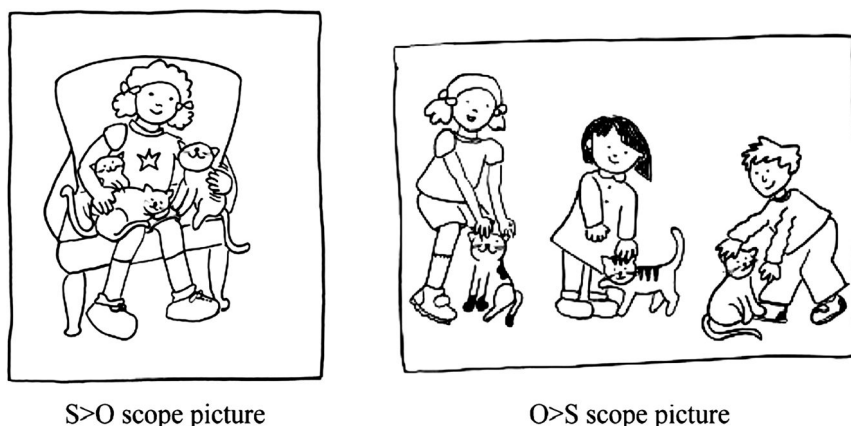


Figure 10.3 Two pictures illustrating TVJT meanings, from Marsden (2009)

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98%, a statistically significant difference. With such high variability in native judgments, Marsden looked carefully at individual results to identify speakers with consistent evaluations. The most significant finding was that half of the advanced English learners consistently rejected the object high scope of Japanese sentences as in (21a), as expected. Since POS by definition entails that there is no overt evidence for the knowledge acquired (unavailability of some interpretation), these learners' knowledge must arise from internal mechanisms, namely, Universal Grammar. The careful design of this study provides convincing evidence that UG is accessible in second language acquisition, even in investigating interpretations that are shaky and prone to individual variation in the learners' native language.

Practice: Before going any further, discuss the differences between syntax–semantics mismatches and POS learning situations. Is there any indication of which one is more difficult to acquire?

10.5 Meaning of novel constructions

This section will present another learning situation studied in second language acquisition, mainly in the work of Laurent Dekydtspotter, Rex Sprouse, and their students at Indiana University. When the second language makes available a syntactic construction whose equivalent does not exist in the native language, learners' first task is to acquire the syntax of the new construction. However, after the construction is learned, all the semantic effects that follow from it will be part of the learners' grammar. As an example, we shall examine interpretations of discontinuous interrogatives, as studied by Dekydtspotter and Sprouse (2001). Consider the examples in (22) and (23) and compare the *wh*-phrases in them. In (22), the whole *wh*-phrase has moved to the beginning of the clause, but the one in (23) has split: the qualifying PP *de célèbre* is left in the verb phrase and you see it below the verb. This one is the discontinuous interrogative.

- (22) *Qui de célèbre fumait au bistro dans les années 70?*

Who of famous smoked in the bar in the 70s?

'Which famous person smoked in bars in the 70s?'

(23) *Qui fumait de célèbre au bistro dans les années 70?*

Who smoked of famous in the bar in the 70s?

‘Which famous person smoked in bars in the 70s?’

Both of these sentences are possible sentences in French; however, their meaning is not exactly identical. A possible answer to the question in (22) may involve a person who is currently famous but was not famous in the 70s, say, the US media celebrity Oprah Winfrey, or a person who was a celebrity in the 70s, say, the soul and blues singer Al Green.⁴ On the other hand, it is impossible to answer the discontinuous interrogative as in (23) with a present celebrity. Only someone who was a celebrity in the past is the appropriate answer.

Without going into the details of the syntactic analysis, the two interpretations depend on where the adjectival phrase *de célèbre* is interpreted in the structure: below or above the tense morpheme. In (22) the adjective is above the tense morpheme on the verb, so a past or present celebrity would fit the bill. In (23) the adjective is below the past tense marker in the structure (in its power, so to say), and only someone who was famous in the past would be the appropriate answer.

What kind of knowledge must an L2 learner have in order to be aware of both interpretations in the case of continuous *wh*-phrases but only one interpretation in the case of discontinuous ones? First, knowledge of overt *wh*-movement is required. It relies on properties of *wh*-words encoded in the functional lexicon, but such knowledge can be transferred from the native language in English-French interlanguage, since both English and French exhibit *wh*-movement. Second, knowledge that discontinuous interrogatives are allowed in French is necessary. This property is not taught in French classrooms (Dekydtspotter and Sprouse 2001) but was given to the participants of the experiment in the form of the test sentences (assuming they did not believe that the researchers had tricked them into judging ungrammatical sentences). Third, the (not taught) language-specific knowledge that French allows the *wh*-word *qui* ‘who’ to have an adjectival restriction at

⁴ I am aware that this example depends on time-sensitive cultural knowledge. For the purposes of this example, assume that Oprah Winfrey was not famous in the 70s but is now famous as an actor as well as a media personality; Al Green was a famous soul and blues star in the 70s although young people may not have heard of him today. Both are amazing artists, as viewing their performances on YouTube can ascertain.

all is necessary, while English *who famous* and *who of famous* are not legitimate phrases. Most importantly, however, what Dekydtspotter and Sprouse labeled *the universal deductive procedure* is indispensable for reaching the interpretive knowledge. The authors make a convincing case for the interpretations not being learnable on the basis of input alone and not transferable from English, making this another POS situation.

The task of the participants was to read a paragraph-length context in English (the native language of the learners) matched with a test question-and-answer pair in French, the target language. After each story and Q–A pair, the participants had to indicate whether that was the correct answer to the question in the context of the story. They were not asked to judge the acceptability of the question strings, just the appropriateness of the answer. Dekydtspotter and Sprouse were the pioneers this type of design. I have changed the example story a little to accord with my examples of famous people.

- (24) Example context and test quadruple, after Dekydtspotter and Sprouse (2001)

Attitudes toward smoking have changed drastically since the 1970s. In the 70s, many people would go to bars and smoke every night. For example, Al Green was a famous star in those days and was often seen at bars smoking with Oprah Winfrey, who was then totally unknown. How times have changed! Now it is Oprah Winfrey who is famous, and neither of them smokes anymore!

Continuous interrogative with past time answer:

Mme Goyette: *Qui de célèbre fumait – dans le bistro – pendant les années 70?*

Élève: *Al Green.*

Continuous interrogative with speech time answer:

Mme Goyette: *Qui de célèbre fumait – dans le bistro – pendant les années 70?*

Élève: *Oprah Winfrey.*

Discontinuous interrogative with past time answer:

Mme Goyette: *Qui fumait de célèbre – dans le bistro – pendant les années 70?*

Élève: *Al Green.*

Discontinuous interrogative with speech time answer:

Mme Goyette: *Qui fumait de célèbre – dans le bistro – pendant les années 70?*

Élève: *Oprah Winfrey.*

Question for respondents on all items:

Is this a correct answer to the question?

The results reveal that past-time readings were preferred for continuous and discontinuous interrogatives by natives and learners alike. In this respect, learners followed the native pattern. It is knowledge of the unavailable interpretation, namely, the speech-time reading (in example (23), someone who is a celebrity at present) with discontinuous interrogatives that is crucial in answering the research question of this study. Both intermediate and advanced learners demonstrated a statistically significant difference in their acceptance of the available and the unavailable interpretations, in the expected direction. In other words, they reliably treated the two constructions differently.

Learners were capable of successfully combining the properties related to the French functional lexicon—the availability of *wh*-movement and discontinuous interrogatives—with the universal meaning-calculating algorithm. Even not very proficient L2 learners, in this case learners with as little as three semesters of exposure to French, manifested knowledge depending on this universal algorithm. It is no small achievement on the part of the learners that even with infrequent strings or subtle interpretations and under severe POS, they manage to exhibit the contrasts we expect based on the respective syntactic structures and the universal meaning computation procedure. Let me reiterate that meanings that obtain as a result of two possible derivations of the syntactic construction, such as present and past readings of the adjectival restriction (a present celebrity and a past celebrity) follow from the normal compositional-semantic calculation of the two available structures. Once learners are aware of the syntactic structures, the two meanings come for free. The case will be made in this book that in such learning situations, it is the syntax that has to come first while the semantics will inevitably follow, hence there is no need to teach it (see Chapter 13).

10.6 Conjuring up something from nothing

A special type of learning situation concerning meaning is the case when a grammatical conceptual meaning such as definiteness is captured by dedicated morphemes in the L2 but presented by a variety of means, including lexical items, aspectual prefixes, word order, and context in the native language. This particularly difficult task will be illustrated with the work of Ionin, Ko, and Wexler (2004) on Russian and Korean native speakers acquiring definite articles in English. These authors argue that the acquisition of articles is regulated by a semantic parameter, and that the patterns of learner accuracy in definite and specific contexts support the conclusion that learners are setting a parameter value. Another way of conceptualizing the learning task in Russian-English interlanguage is to think of one-to-many mapping: there is one way of signaling definiteness in English but many ways in Russian, including fixing definiteness values through the context.

It is well known that speakers of a language without articles (e.g., Japanese, Chinese, Korean, Russian) have a hard time using articles correctly in English. A number of experimental studies⁵ have identified two types of errors these learners make: they either omit articles altogether or use them in inappropriate contexts. While these studies identified the error pattern in definite article use and linked it to the specificity of the nominal phrase in the target language, they offered no principled explanation as to why L2 learners' choices should be affected by specificity. A series of recent studies, starting from Ionin (2003) and Ionin, Ko, and Wexler (2004), provide a principled, parameter-based explanation of learner article interpretation and use. Before looking at the learning task and the syntax-semantics mismatch, let us go over some informal definitions of definiteness and specificity, following Ionin, Ko, and Wexler (2004).

Definiteness and specificity are semantic features; both are part of the conceptual arsenal of language. They are also discourse-related since they rely on the knowledge of the speaker and the hearer in a communication situation. By using a definite nominal phrase, a speaker refers to a uniquely identified individual in the mind of the speaker and the hearer. Definiteness is morphologically encoded by *the*; indefiniteness is encoded by *a(n)* for

⁵ E.g., Huebner (1983), Robertson (2000), Thomas (1989).

singular count nouns and by the zero article for plural nouns (e.g., *a car-cars*). Uniqueness can be established either by prior mention, or by shared knowledge. Take (25) for example.

- (25) Julie saw a car in the driveway. The car was red and shiny.

Since the first mention of the object (the car) does not satisfy the presupposition of uniqueness in the mind of the speaker and the hearer, it is indefinite and marked with *a*. At the second mention, the object is already established in the discourse as known to the speaker and the hearer, hence it is definite and marked with *the*. An object need not be mentioned in the discourse immediately preceding the definite noun phrase if it is known to both the speaker and the hearer. The example in (26) is an out-of-the-blue statement between a husband and wife who have ordered a computer repair person to visit their home:

- (26) The computer guy is coming at 10. Can you stay at home to meet him?

Specificity also reflects the property of uniqueness, an object or person being activated in the mind, but only of the speaker, not of the hearer. A specific nominal phrase has to be noteworthy in some discourse-related way. Specificity is not marked morphologically in standard English; that is, both indefinite and definite nominal phrases can be specific or not, as the examples in (27) illustrate. Examples (27c,d) are from Ionin, Ko, and Wexler (2004), their (9a,b).

- (27) a. Jill wants to marry a Canadian. She is going to present him to her family at Christmas. (indefinite specific)
b. Jill wants to marry a Canadian, but she has not met one yet. (indefinite nonspecific)
c. I want to speak to the winner of today's race—she is my best friend! (definite specific)
d. I want to speak to the winner of today's race—whoever that is; I'm writing a story about this race for the newspaper. (definite nonspecific)

Colloquial English does have a marker of specificity, the demonstrative *this*, but its usage is optional and colloquial, see (28).

- (28) I saw a/this kitty in the pet shop, and I want to buy her for my daughter.

When speakers of a language without articles have to acquire articles in English, their learning task is to map the semantic feature definiteness onto its morphological expression *the*, and the lack of it to *a(n)* in singulars or zero article in plurals. Note that in article-less languages definiteness is established through context and other linguistic means such as demonstrative pronouns but does not have dedicated morphology. Thus, the semantic feature definiteness is not new to the learners, but its morphological expression is new. The two semantic features, specificity and definiteness, are somewhat close in meaning, having to do with establishing uniqueness of an object in the discourse. It is not inconceivable, then, for learners to use both features in semantically bootstrapping themselves into the morphology. However, Ionin (2003) and Ionin, Ko, and Wexler (2004) go one step further and propose the Article Choice Parameter as in (29), a principled explanation of how languages choose which feature their articles reflect.

- (29) A language that has two articles distinguishes them as follows:
The definiteness setting: Articles are distinguished on the basis of definiteness (exemplified in English)
The specificity setting: Articles are distinguished on the basis of specificity (exemplified in Samoan)

Cross-linguistic evidence for this semantic parameter comes mainly from Samoan, a language that purportedly uses the article *le* with specific nominal phrases and the article *se* with nonspecific ones, regardless of definiteness. Based on the Article Choice Parameter, Ionin, Ko, and Wexler (2004) predicted that learners of English from article-less languages would fluctuate between the different settings of the UG-supplied semantic parameter until the input leads them to the target value (definiteness). Their native language would not aid them in this choice since it distinguishes both features, but none of them morphologically. Table 10.4 summarizes these predictions.

Ionin, Ko, and Wexler tested beginning, intermediate, and advanced learners of English with Russian or Korean as their native languages. They employed a forced-choice elicitation task and a production task. The

Table 10.4 Predictions for article choice in L2 English by learners whose L1 lacks articles

	Definite NP (Target <i>the</i>)	Indefinite NP (Target <i>a</i>)
Specific NP	Correct use of <i>the</i>	Overuse of <i>the</i>
Nonspecific NP	Overuse of <i>a</i>	Correct use of <i>a</i>

forced-choice elicitation task included short dialogs for context, the test sentences were part of the dialogs as in (30).

(30) Definite nonspecific context:

Conversation between a police officer and a reporter:

Reporter: *Several days ago, Mr. James Peterson, a famous politician, was murdered! Are you investigating his murder?*

Police officer: *Yes, we are trying to find (a, the, __) murderer of Mr. Peterson, but we still don't know who he is.*

We will look at the results of the elicitation task from the intermediate and advanced Russian learners only. The Korean learners were much more accurate than the Russian learners and did not exhibit the expected fluctuation pattern as clearly as the Russians, since they were performing at ceiling. Table 10.5, adapted from Ionin, Ko, and Wexler's Table 12, indicates the Russian group results. The percentages in the cells do not add to 100 because article omissions are excluded. Compare Tables 10.4 and 10.5. As the group results in Table 10.5 show, the predictions of the Article Choice Parameter were supported indeed. Learners' behavior was not random, their choice of article was significantly affected by the specificity of the nominal phrase. Compare the choices of articles when the target is definite: the learners choose the definite article more often when the noun phrase is specific. The opposite is true when the target is indefinite.

The Article Choice Parameter and the related fluctuation prediction turn out to be a principled explanation of learners' ostensibly erratic, but actually predictable article choice. In addition, it is an eminently testable hypothesis. Moreover, as more English L2 groups are being tested from different languages without articles, some interesting variation is uncovered. For example, the presence of classifiers in Chinese may be aiding the learners in acquiring English articles, while the presence of demonstrative pronouns (which are specificity and deixis markers) is not aiding Polish and Russian

Table 10.5 Percentage use of articles in different contexts by Russian L2 learners of English

	Definite NP (Target <i>the</i>)	Indefinite NP (Target <i>a</i>)
Specific NP	79% <i>the</i> , 8% <i>a</i>	36% <i>the</i>, 54% <i>a</i>
Nonspecific NP	57% <i>the</i>, 33% <i>a</i>	7% <i>the</i> , 84% <i>a</i>

learners in a similar way. However, the main conclusion remains that, even though the learners cannot transfer any morphology knowledge from their native languages, the universal semantics helps them to bootstrap themselves into language-specific morphological expressions of definiteness. I have to add that there are other accounts (e.g., Trenkic 2008) of L2 article difficulty, but the Article Choice Parameter has gained currency and support from replications.⁶

10.7 Conclusions

We saw several learning situations in this chapter related to the acquisition of some meaning coupled with some form. In the POS situation, two sentences in the target language seem very similar in form; however, one of them is ambiguous while the other is not. By definition, this configuration of form–meaning mappings cannot be acquired without negative evidence. In a syntax–semantics mismatch, similar forms and meanings exist in both the L1 and the L2; however, they are misaligned (see Table 10.1). In principle, positive evidence should be sufficient for acquiring such mismatches, but in practice they create difficulty for learners because the native form–

⁶ Ionin, Zubizarreta, and Philippov (2009) modified the original Fluctuation Hypothesis proposed in Ionin et al. (2004). New linguistic research (Fuli 2007, Tryzna 2009) has showed that instead of distinguishing between articles based on specificity and not on definiteness, Samoan distinguishes specificity, but within indefinites only. Since the universal specificity distinction is only demonstrated within a part of the article space, children and adult L2 learners are expected to be overusing *the* with specific indefinites, but not overuse *a* with nonspecific definites. Ionin et al. (2009) argue that only specificity-related errors with indefinites, not specificity-related errors with definites, reflect L2-learners' access to the semantic universal of specificity. Their revised proposal was anticipated in some of the results from Ionin et al. (2004) and receives support from further findings with L1-Russian children acquiring English.

meaning mapping is securely entrenched. Thirdly, in situations where the syntax maps onto interpretations in a regular fashion (while both are unavailable in the L1), the universal semantic computation mechanism can be employed and acquisition does not seem problematic. Finally, functional elements such as articles that can potentially encode two meanings cross-linguistically (definiteness and specificity) are a big challenge. If the native language does not encode either of these meanings with overt and dedicated morphology, then choosing between the two meanings for L2 articles may take considerable exposure to input and practice. I should add that article meaning is acknowledged as an extremely difficult area of L2 acquisition, possibly because article-less languages have other means of expressing specificity and definiteness (such as topic and focus, word order, demonstratives, etc.). If we look at the learning task in this way, acquiring article semantics turns into a syntax–semantics mismatch and involves (semantic) feature reassembly.⁷

10.8 Exercises

Exercise 10.1. Describe the differences represented in the examples below as syntax–semantics mismatches:

Temporal morphology in Italian and Portuguese (Giorgi and Pianesi 1997: 50)

- | | | | |
|------|----|---|------------|
| (i) | a. | <i>Ho mangiato alle quattro.</i> | Italian |
| | | have.SG eaten at four | |
| | | ‘I have eaten at four (e.g., once before/all my life).’ | |
| | b. | <i>Mangiai alle quattro.</i> | |
| | | ate.SG at four | |
| | | ‘I ate at four.’ | |
| (ii) | a. | <i>Comi as quatro.</i> | Portuguese |
| | | ate.SG at four | |
| | | ‘I have eaten at four (e.g., once before/all my life) / I ate at four.’ | |

⁷ See Cho and Slabakova (2014) for discussion.

- b. *Tenho comido as quatro.*
 have.SG eaten at four
 ‘I took the habit of eating at four’

How would you describe this mismatch? Pay attention to the forms of the main verb and the auxiliary *have* on the second line of the examples, the gloss, and then compare the meanings given in the translations on the third line of each example.

Tense in Mandarin Chinese and English:

- (iii) *Zhang San zhù zài zhèr.*
 Zhang San live at here
 ‘Zhang San lives here.’
- (iv) *Zhang San 1989 nián zhù zài zhèr.*
 Zhang San 1989 year live at here
 ‘Zhang San lived here in 1989.’

Now examine another mismatch as in (iii) and (iv) and compare the Mandarin verb form with the English verb form in the translations of the Chinese examples. How do we know which Chinese sentence describes a past event, if we see/hear these sentences out of context?

Finally, compare the Italian–Portuguese mismatch and the Mandarin–English mismatch. Which one do you think would present more difficulty for second language learners, and in which learning direction?

Exercise 10.2. Examine the last two groups of columns in Figure 10.1. They reflect the mean ratings of participants on sentences such as *I observed Matt eat a cake* and *I observed Matt eating a cake* in the context of a complete story (Matt finished the cake). In principle, both sentences should be judged as plausible, since to observe an event in its completion includes the observation of the process as well as the end of the event. Compare the ratings of the native speakers for the two test sentences in the complete context. What do they show? Is the pattern repeated in any other group of participants?

Now, recall that the native language of the learners has only one form for *eat* and *eating* in this type of sentence; that is, Bulgarian does not distinguish between the bare verb and the present participle in perceptual reports. How would you evaluate the learners’ performance on the English contrast,

keeping in mind the native language? Discuss the performance of all learner groups. Remember that chance is 50% in this experimental design.

Exercise 10.3. Look at the Ionin, Ko, and Wexler (2004) experimental setup again. Below is another context and test sentence (forced choice) from that test.

Indefinite specific context:

Phone conversation:

Jeweler: *Hello, this is Robertson's Jewelry. What can I do for you, ma'am?
Are you looking for some new jewelry?*

Client: *Not quite. I heard that you also buy back people's old jewelry.*

Jeweler: *That is correct.*

Client: *In that case, I would like to sell you (a, the, ____) beautiful silver necklace. It is very valuable, it has been in my family for 100 years!*

Come up with two other context–sentence pairs: a definite specific context and an indefinite nonspecific context. Make them plausible and easy to understand by nonnative speakers.

In Ionin, Zubizarreta, and Philipov (2009), the authors use a very similar design but they make one important change in the elicitation task: they do not provide articles to choose from in the test sentence (*a, the, ____*) but leave the space blank and ask the participants to provide the appropriate word, including absence of any word. The test sentence would look like that:

Client: *In that case, I would like to sell you () beautiful silver necklace. It is very valuable, it has been in my family for 100 years!*

Which task is harder and why? The elicitation task has been criticized as possibly tapping extralinguistic, or learned, knowledge of grammar, especially with classroom learners. What other task can be used for the elicitation of articles that would avoid this criticism?

Exercise 10.4. Still staying within the article choice parameter, in another study, Ionin, Zubizarreta, and Maldonado (2008) compared Russian learners of English and Spanish learners of English on the same task. What is the motivation for such an experimental design, if Spanish and English both have overt articles while Russian does not? Tables 10.6 and 10.7 below give the main results. Comment on what these results show.

Table 10.6 Elicitation test results: L1-Russian speakers (n = 19), Ionin et al. (2008: 564)

	Definite NP (Target <i>the</i>)	Indefinite NP (Target <i>a</i>)
Specific NP	93% <i>the</i>	23.7% <i>the</i>
	5.3% <i>a</i>	74.6% <i>a</i>
	0% __	0% __
	0.9% other	1.8% other
Nonspecific NP	86% <i>the</i>	3.5% <i>the</i>
	14% <i>a</i>	95.6% <i>a</i>
	0% __	0% __
	0% other	0.9% other

Table 10.7 Elicitation test results: L1-Spanish speakers (n = 20), Ionin et al. (2008: 565)

	Definite NP (Target <i>the</i>)	Indefinite NP (Target <i>a</i>)
Specific NP	87.5% <i>the</i>*	1.7% <i>the</i>
	0.8% <i>a</i>	92.5% <i>a</i>
	8.3% __	0% __
	2.5% other	5% other
Nonspecific NP	96.7% <i>the</i>	4.1% <i>the</i>
	0.8% <i>a</i>	91.7% <i>a</i>
	1.7% __	0.8% __
	0.8% other	3.3% other

*Note that the unexpected lower accuracy on this category of items was traced to only one example that was a direct transfer from the native language, since Spanish requires article omission in this context (*Fui a casa*, lit. ‘I went to house’).

Do you think these findings offer evidence for the Article Choice Parameter and why/why not?

Exercise 10.5. We shall look at a complete study, Gabriele (2005), which tests a Japanese–English contrast within aspectual tenses. I shall describe the syntax–semantics mismatch to be acquired, the research questions and the study, and ask some questions along the way. At the end, you will be asked to interpret the results and draw conclusions.

On the surface, it looks like the Japanese inflectional morpheme *te-iru* and the English progressive form *be V-ing* are a perfect match. With most event verbs (ii-a) and all incomplete activities (ii-b), *te-iru* has an ongoing interpretation, as does *be V-ing* (i-a,b). The mismatch is manifested only with instantaneous event verbs: while *be V-ing* is unnatural with some such verbs (iii-a), with some others it highlights the process immediately preceding the change of state (iii-b). Crucially, *te-iru* cannot have an ongoing interpretation but only a completion, that is, a perfective interpretation (iv). The Japanese examples are from Gabriele (2005).

- (i) a. *Samantha is making a cake.*
 b. *Samantha is running.*
- (ii) a. *Ken-wa* *isu-o* *tukut-te i-ru*
 Ken-TOP chair-ACC make-ASP-NONPAST
 ‘Ken is making a chair.’
- b. *Ken-ga* *utat-te i-ru*
 Ken-NOM sing-ASP-NONPAST
 ‘Ken is singing.’
- (iii) a. **I am losing my wallet.*
 b. *The plane is arriving at the airport.*
- (iv) *Hikōki-ga* *kūkō-ni* *tui-te i-ru*
 plane-NOM airport at arrive-ASP-NONPAST
 The plane has arrived at the airport.
 DOES NOT MEAN ‘The plane is arriving at the airport.’

The meaning of Japanese *te-iru* interacts with aspectual verb class, in the sense that one meaning (ongoing) is available for some verbs (events and activities) but another meaning (perfective) surfaces with other verbs (instantaneous events). Does this meaning difference between Japanese and English constitute a syntax–semantics mismatch and why, or why not?

Gabriele (2005) presents two studies of this syntax–semantics mismatch, one of Japanese as a second language and one of English as a second language. She had somewhat different research questions and employed complex tests with many conditions, but here we will only concentrate on the results relevant to the acquisition of the present progressive *be V-ing* and

present *te-iru* (the past form is *te-ita*). In order to test sentence interpretation, she used a modified Truth Value Judgment Task with pictures. In the two studies, the stories were presented aurally in the target language, not in the native language. Learners listened to recorded stories and saw two pictures. They were then presented with a sentence both visually and aurally and asked to judge on a scale from 1 (worst) to 5 (best) whether the sentence made sense as a description of the story. For each verb, a complete context and an incomplete context were provided. Here are some examples:

- (v) a. Complete event context for *paint a portrait*, an event VP
 Picture 1: Ken is an artist. At 12:00 he begins to paint a portrait of his family.
 Picture 2: At 8:00 he gives the portrait to his mother for her birthday.
Ken is painting a portrait of his family.
 (Expected answer 1 in Japanese and English)
- b. Incomplete event context for *paint a portrait*, an event VP
 Picture 1: Ken is an artist. At 12:00 he begins to paint a portrait of his family.
 Picture 2: At 12:30 he paints his mother and father.
Ken is painting a portrait of his family.
 (Expected answer 5 in Japanese and English)
- (vi) a. Complete event context for *arrive*, an instantaneous event VP
 Picture 1: This is the plane to Tokyo. At 4:00 the plane is near the airport.
 Picture 2: At 5:00 the passengers are at the airport.
The plane is arriving at the airport.
 (Expected answer 1 in English, 5 in Japanese)
- b. Incomplete event context for *arrive*, an instantaneous event VP
 Picture 1: This is the plane to Tokyo. At 4:00 the plane is near the airport.
 Picture 2: There is a lot of wind. At 4:30 the plane is still in the air.
The plane is arriving at the airport.
 (Expected answer 5 in English, 1 in Japanese)

In study 1, Gabriele tested 101 native speakers of Japanese learning English in Japan and 9 near-native Japanese individuals living in New York City, as well as 23 native Japanese as controls. In study 2, she tested 33 native English speakers learning Japanese in the US, as well as 31 native speaker controls. Proficiency tests were used to divide the participants into groups. One thing to note is that the Japanese L2 low group is of lower proficiency than the English L2 lowest group.

Turning to results, we shall look at the Japanese → English direction first (Table 10.8).

Compare the two ratings within groups, starting with the native speakers. If a group has a sufficient distance in its ratings of the acceptable and

Table 10.8 Mean scores on instantaneous events in the English present progressive (range between 1 and 5)

	Low (n=46)	Interm (n=39)	High (n=16)	Near-native (n=9)	Native (n=23)
<i>is arriving</i> refers to incomplete event	3.08	2.89	4	4.25	4.1
<i>is arriving</i> refers to complete event	3.64	3.45	2.56	3.25	1.5

Table 10.9 Mean scores on instantaneous events with *te-iru* (range between 1 and 5)

	Low (n=16)	High (n=17)	Native (n=31)
<i>tui-te i-ru</i> ‘is arriving’ refers to incomplete event	3.28	2.04	1.35
<i>tui-te i-ru</i> ‘is arriving’ refers to complete event	4.39	3.93	4.84

unacceptable meanings, this suggests that the group accepts the appropriate meaning and also rejects the unavailable one. Answer the following questions on the results. Have the four groups of Japanese learners of English acquired the incomplete event interpretation? Have they acquired the fact that *The plane is arriving* does not mean *The plane has arrived*? Which of the four groups seems to be doing the best? (Statistical analysis is necessary to answer this question, but estimate the answer by just eyeballing the ratings.)

Next, look at the results of the English → Japanese direction (Table 10.9). What can we say about this learning direction? Have the English learners of Japanese acquired the complete event interpretation of *te-iru*? Have all levels acquired it? Is there a difference between the two learning directions, English → Japanese and Japanese → English? Describe the difference, using comparisons between equivalent levels of proficiency in the L2. Can you speculate on why this would be the case?

Now compare the accuracy on the constructions that are the same in the native language and the accuracy on constructions that have to be added to the L2 grammar, the ones that cannot be transferred from the native grammar. Is there a pattern of (successful or unsuccessful) acquisition that you see? Describe the patterns in this way: Acquiring an L2 construction or meaning that also exists in the native language is... Acquiring an L2 construction or meaning that also is new in the second language is...